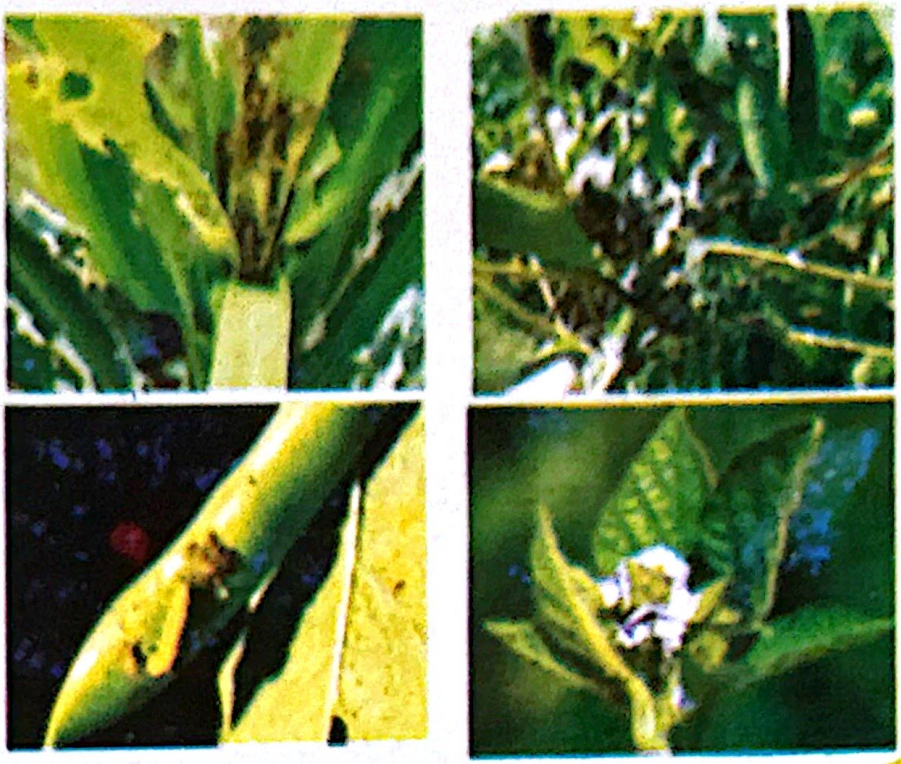
GUIDE ON

CROP PEST AND DISEASES MANAGEMENT



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GUIDE ON CROPS PESTS AND DISEASES MANAGEMENT

Overview

This guide provides practical information on identifying and managing common pests and diseases affecting various crops.

Introduction of pests and pathogens

Pests are harmful creatures that affects plants and their produce. These are insects, rodents, and weeds. Pathogens however are disease-causing agents that infect plants either directly or through a carrier known as a vector. They are in various forms and are classified as bacteria, fungi, viruses, and nematodes.

Common symptoms to look out for

1. INSECT INFESTATIONS

These are usually seen as holes in leaves (a) and pods (b), flower or fruit deformities (c) and leaf distortion (d).






FUNGAL INFECTIONS

Rust disease: Leaves often look reddish-brown like rusted iron and may twist or show distortion. The disease can affect the vigour and yield of plants (e).

Leaf spot diseases: Typical leaf spot symptoms include light-coloured, circular spots with light-brown edges. Heavy infection leads to leaf loss (f).

Leaf Blight: Small yellowish round to oval spots is seen on the leaves. The spots gradually increase in area into bigger elliptical spots and are straw to grayish brown in the centre with dark brown margins. The spots coalesce giving blighted appearance (g).

Powdery mildew: Symptoms consist of a light, greyish, powdery growth on the leaves, pods and occasionally the stems. This powdery growth is easily rubbed off. When the disease is severe, plants turn yellow and defoliate (h).






3. VIRAL INFECTIONS

Leaf blistering (i), chlorosis, curling and vein banding (j): Single or combinations of these symptoms may signify viral infection. Late response can cause stunting and reduced yield.



4. NEMATODE INFECTIONS

Nematodes (k) invade plant root systems, forming root galls or knots (l) which disrupt water and nutrient absorption, leading to stunted growth, yellowing, and lower yields. Symptoms include wilting and excessive root branching.



5. WEEDS INFESTATIONS

Weeds cause different issues on farms based on the types of crops being grown. They compete with and sometimes deprive main crops for water, sunlight, and nutrients, reducing crop yields. Weeds with similar leaf morphology as crops are sometimes difficult to differentiate which makes weeding process more challenging. Weeds also harbor pests and pathogens that attack crops.

Examples of common weeds in crops are Spurge (*Euphorbia heterophylla*) (m) and Spear grass (*Imperata cylindrica*) (n)



PEST AND DISEASE MANAGEMENT

Pests and diseases can be managed in two broad ways either as a preventive (before attack) or control (after attack) measures.

A. Preventive Measures

This involves steps taken before attack to create an unfavorable environment for pests/pathogens, reducing the likelihood of attack. To achieve this, farmers should consider several key steps. Understanding the cropping history of the selected site is crucial to anticipating potential pest and disease problems. It is advisable to avoid sites known to harbor pests targeting the crop of interest. Sourcing pest/disease resistant planting materials from trusted sources prevents severe damage. Pre-planting treatments, such as seed treatments for crops prone to early seedling infestations/infections are also effective. Additionally, planning planting times to avoid peak pest/pathogen and weed seasons helps prevent infestations.

B. Control Measures

Chemical method: Bordeaux mixture or the use of copper-based fungicide is highly effective for most fungal diseases. For insects, suitable chemicals following manufacturer's recommendation should be applied. Some common insecticides are cypermethrin and lambdacyhalothrin. Weeds control may be achieved using chemicals, applied before (pre-emergence) or after (post-emergence) crop growth. These chemicals work either systemically (internal) or through contact (surface). In the case of nematode infestation, nematicides can be applied to control nematode populations. However, their use should be managed carefully to avoid environmental harm and resistance build up.

Physical Method: Soil solarization (using plastic sheets to trap solar heat), and mechanical removal of infected plants (rogueing) can help physically reduce nematodes, insects, fungi and weeds populations. This is especially effective in small-scale farming.

Biological Method: The use of non-predatory natural enemies of pests/pathogens as well as use of biopesticides they are often used to control insect, nematode, weed population and disease incidence. This method is eco-friendly and very effective.

Cultural Methods: Crop rotation and intercropping practices may be used to disrupt pest and pathogen life cycles and enhance soil health. Use of cover crops and mulches helps shade out weeds and reduce their growth.

Integrated Pest/Disease Management:

This combines different control strategies to improve long-term results. Instead of relying on one method, farmers can use multiple approaches to tackle pests and diseases at differe nt stages of their growth cycle. This not only ensures better control but also reduces negative environmental impacts and chemical resistance.

Conclusion

Sustainable crop production and food security is enhanced when farmers embrace a holistic approach of pest and disease management that combines both preventive and control measures at all times.

Our Vision

A World Class Institute of Agricultural Research, responsive to farmers' and other stakeholders' needs towards Sustainable Development

Our Mission

To develop, validate and transfer appropriate demand-driven agricultural technologies that are ecofriendly, affordable and adaptable through effective agricultural research and extension system.

Core Values

Excellence, Team work, Discipline, Innovation, Transparency, Integrity.

Motto

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